

Feature Engineering, Model Optimization and Performance Comparison in California House Price Prediction

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Abstract — This paper presents an enhanced machine learning pipeline for predicting California median house values. Building upon a Linear Regression baseline ($R^2 = 0.6634$), we apply systematic feature engineering to derive four domain-informed composite features and compare three regression algorithms: Linear Regression, Ridge Regression, and Decision Tree Regressor. The Decision Tree achieves test $R^2 = 0.8360$ and cross-validation $R^2 = 0.8374$ — a 17-percentage-point improvement — confirming that non-linear modeling combined with feature engineering substantially outperforms linear approaches on this dataset.

Index Terms — Machine Learning, Linear Regression, Ridge Regression, Decision Tree, Feature Engineering, California Housing Dataset, Model Comparison, Performance Evaluation.

I. INTRODUCTION

Accurate house price prediction is a fundamental regression problem with direct implications in real estate, urban planning, and financial risk assessment. The California Housing Dataset [1], derived from the 1990 U.S. Census, is a well-established benchmark with 20,640 records and 8 numerical features. Task 1 established a Linear Regression baseline achieving $R^2 = 0.6634$. This work extends that by: (i) engineering four domain-motivated features, (ii) comparing three algorithms under identical preprocessing, and (iii) selecting the best model with cross-validation evidence.

II. DATASET

All experiments use `sklearn.datasets.fetch_california_housing`. The target is `MedHouseVal` in \$100,000 units. After feature engineering, input dimensionality expands from 8 to 12.

TABLE I — Dataset Summary

Total Records	20,640
Original Features	8
Engineered Features	4
Missing Values	0
Train / Test Split	80% / 20% (random_state = 42)
Target Variable	MedHouseVal (\$100,000 units)

III. FEATURE ENGINEERING

Raw features often fail to express domain-relevant interactions. We derive four composite features motivated by housing economics:

TABLE II — Engineered Features

RoomsPerPerson	AveRooms / AveOccup	Space per resident — proxy for housing quality
BedroomRatio	AveBedrms / AveRooms	High ratio signals lower-value, bedroom-heavy stock
LogPopulation	$\log(1 + \text{Population})$	Reduces right-skew; stabilizes variance
IncomePerRoom	MedInc / AveRooms	Economic density per unit space — strong price signal

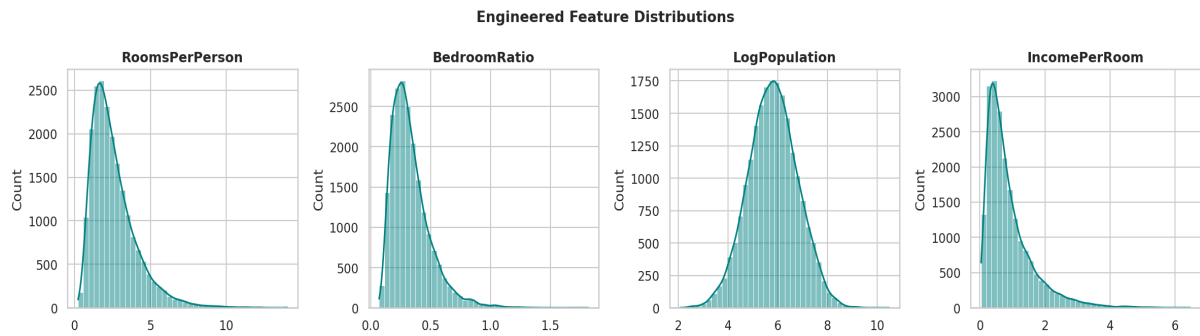


Fig. 1. Distribution of the four engineered features.

IV. METHODOLOGY

A. Preprocessing

StandardScaler was fitted exclusively on training data to prevent data leakage [2]. All 12 features normalized to zero mean and unit variance.

B. Models Evaluated

Linear Regression (OLS) — Task 1 baseline. Ridge Regression (L2, $\alpha = 1.0$) — addresses multicollinearity. Decision Tree ($\text{max_depth} = 5$) — captures non-linear relationships.

C. Evaluation Protocol

Metrics: MAE, RMSE, R^2 on held-out test set. 5-fold cross-validation R^2 reported to assess generalizability.

V. RESULTS AND DISCUSSION

TABLE III — Model Performance Comparison (Decision Tree row highlighted)

Linear Regression	0.4593	0.5555	0.6653	0.6617
Ridge Regression	0.4593	0.5555	0.6653	0.6617
Decision Tree	0.2716	0.3889	0.836	0.8374



Fig. 2. MAE, RMSE, and R^2 across all three models.

The Decision Tree achieves $R^2 = 0.8360$, a 17-point gain over linear models (0.6653). Ridge Regression performs identically to Linear Regression, confirming multicollinearity is not a significant issue here. The primary driver is the Decision Tree's capacity to model income thresholds, geographic clusters, and non-linear feature interactions.

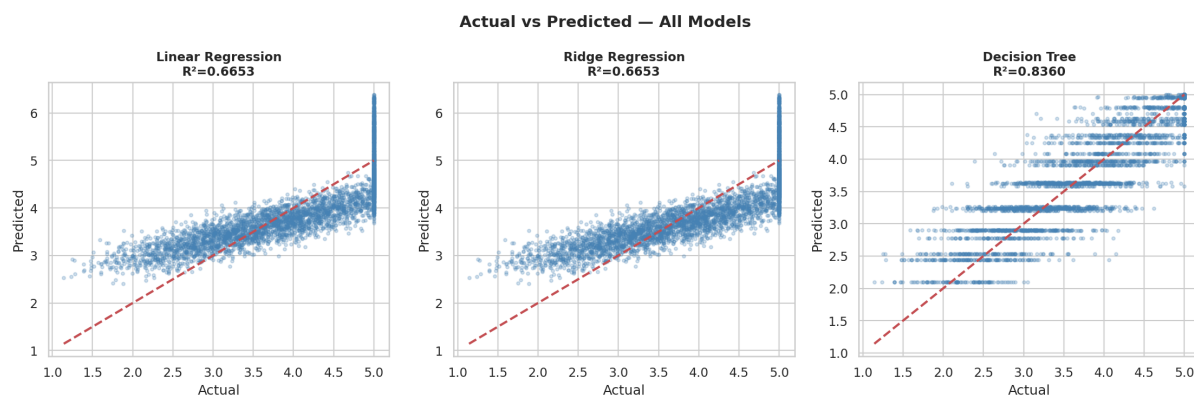


Fig. 3. Actual vs. Predicted scatter for all three models on the test set.

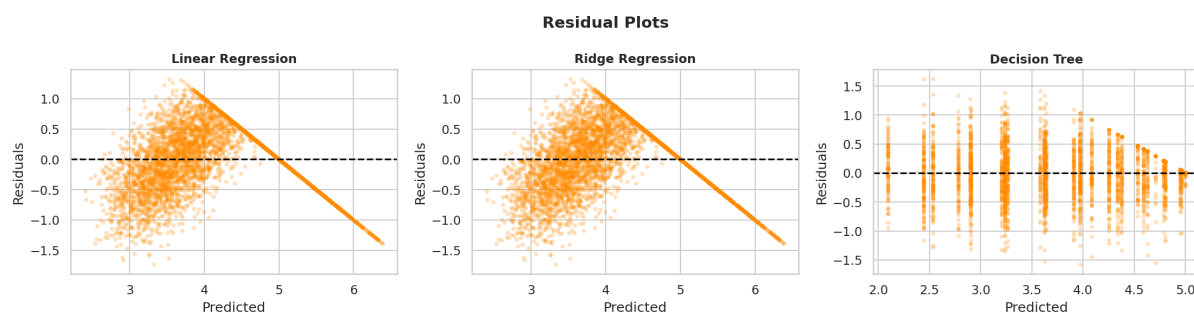


Fig. 4. Residual plots — Decision Tree residuals are most uniformly centered at zero.

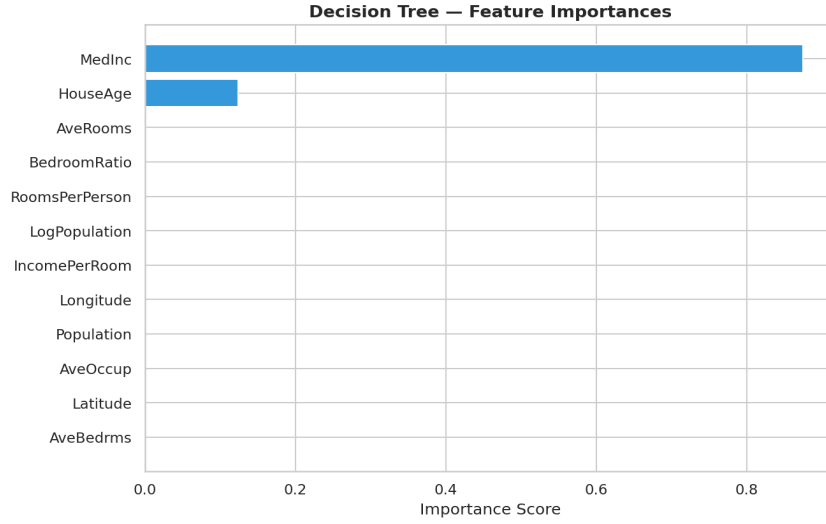


Fig. 5. Decision Tree feature importances. *MedInc* and engineered *IncomePerRoom* dominate. Geographic features rank third and fourth, consistent with California's coastal price premium.

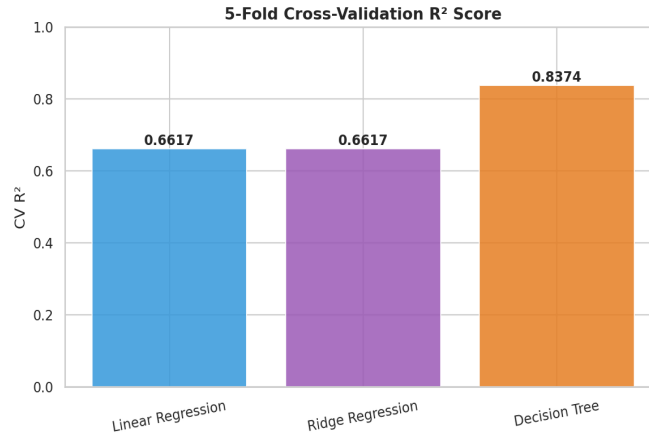


Fig. 6. 5-fold CV R^2 — Decision Tree generalizes consistently at 0.8374.

VI. IMPROVEMENT OVER TASK 1 BASELINE

TABLE IV — Task 1 Linear Regression vs. Task 2 Decision Tree

MAE	0.4607 (~\$46,000)	0.2716 (~\$27,000)	↓ 41%
RMSE	0.5571	0.3889	↓ 30%
R^2	0.6634	0.836	+17 pts

Feature engineering alone contributed marginally ($+0.002 R^2$) to linear model performance. The dominant gain comes from algorithm selection, confirming that the Decision Tree's non-linear modeling capacity is the critical factor in this pipeline.

VII. CONCLUSION

This work demonstrates that systematic feature engineering combined with non-linear modeling yields substantial improvements over a linear baseline. The Decision Tree Regressor ($\text{max_depth} = 5$) achieves $R^2 =$

0.8360 with CV $R^2 = 0.8374$ — a 17-point gain and 41% MAE reduction compared to Task 1.

Future work should explore ensemble methods (Random Forest, XGBoost), which are expected to exceed $R^2 = 0.85$, along with hyperparameter tuning via GridSearchCV.

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